

PAPER_615: UQFF Ug 26D Polynomial Defect Expansion

Unified Quantum Field Framework (UQFF)

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Abstract

The gravitational field component U_g is extended from its four-term UQFF expression to incorporate a degree-26 polynomial defect term $\sum_{m=0}^{26} a_m r^m$ representing multi-pole tidal coupling across 26 spatial dimensions. The U_{g4} term is resolved into a 13+13 factorial split yielding the dual BH26 half-hemisphere bound $(13!)^2 = 3.878 \times 10^{19}$.

§1. Introduction

In the original UQFF formulation, $U_g = g \cdot SC_m / UA \cdot (U_{g1} + U_{g2} + U_{g3} + U_{g4})$. The BigBangHypergraphTheory document identifies that the U_{g4} term is itself a 26th-order construct, expressible as a 13+13 differential split, and that the full U_g requires an additional degree-26 polynomial tail for proper 26D embedding.

§2. Expanded U_g Formulation

$$\boxed{U_g = \frac{g \cdot SC_m}{UA} \left(\sum_{i=1}^4 U_{gi} + \sum_{m=0}^{26} a_m r^m \right)}$$

2.1 U_{g4} — The 13+13 Factorial Split

$$U_{g4} = \frac{d^{13}}{dr^{13}}(r \cdot t) \cdot \frac{d^{13}}{dt^{13}}(r \cdot t)$$

$$\bullet \frac{38!}{12!} \cdot \frac{r \cdot t}{r^{39}} \quad \quad$$

The first term: $\frac{d^{13}}{dr^{13}}(r) = 0$ (only r^1 , so $d^{13}r/dr^{13}=0$ for $13>1$)

For the generalized product: $d^{13}/dr^{13}(r \cdot t)$ at order-13 with coupled t :

$$= 13! \cdot t^0 \quad \text{(purely radial factor)} = 13! = 6,227,020,800$$

$$U_{g4} = (13!)^2 + \frac{38!}{12!} \cdot \frac{t}{r^{38}} = 3.878 \times 10^{19} + \frac{38!}{12!} \frac{t}{r^{38}}$$

2.2 Degree-26 Polynomial Tail

$$P_{26}(r) = \sum_{m=0}^{26} a_m r^m$$

where a_m are Vacuum Density Series (VDS) weighting coefficients per radial mode.

§3. Physical Interpretation

The $(13!)^2$ factorial bound corresponds to the dual BH26 horizon:

- Upper hemisphere (dimensions 14–26): 13 radial steps $\rightarrow 13!$ orderings
- Lower hemisphere (dimensions 1–13): 13 temporal steps $\rightarrow 13!$ orderings
- Product: $(13!)^2 = 3.878 \times 10^{19}$ — the maximum irreducibility count for the U_{g4} coupling

§4. VDS / DVP / BH26 Connections

- **VDS**: a_m coefficients index vacuum density occupation per polynomial mode.
- **DVP**: Degree-26 polynomial irreducibility follows DVP prime-gap uniqueness theorem.
- **BH26**: $13! \times 13! = (13!)^2$ corresponds to dual BH26 half-horizon factorial bound.

§5. Conclusions

The expanded U_g with degree-26 polynomial defect and the re-derived U_{g4} $13+13$ split correctly accounts for all 26 tidal modes of the gravitational field in the UQFF embedding space. The $(13!)^2$ bound prevents field degeneracy across the BH26 dual horizon.

Class: [UQFFUg26DPolynomialDefectExpansionCalculator](#) (#202, CP4 v5.17) **Source:** [grok_share_79fdf5367d1.txt](#) (161 lines, March 29, 2026)

